

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 06 · Graphs of Functions

With Answers

Questions with full worked solutions

- 58 pages
- Past papers up to June 2026

Graphs of Functions

Your classified route

19
questions

165
marks

3
families

8
variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Graphs of Functions

Families and variants

F01 • Polynomial graph grammar

V01 Factorised cubic: intercepts, multiplicity, sign and linked algebra

V02 Reconstruct a cubic from graph data, then intersect another curve

V03 Horizontal-line intersections and geometric separation

F02 • Reciprocal graphs and intersection counts

V01 Shifted a/x graph with line-intersection discriminant

V02 Reciprocal-square symmetry and four-intersection parameter range

V03 Count real roots by comparing polynomial and reciprocal sketches

F03 • Curves, intersections, regions and models

V01 Line-quadratic intersections and region inequalities

V02 Compare contextual quadratic and linear models with a valid domain

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Polynomial graph grammar

WHAT THIS FAMILY ASKS

Use factorised polynomial structure, root multiplicity, intercepts, end behaviour and algebraic form to sketch, read or reconstruct cubic graphs.

VARIANT ROUTE

V01 Factorised cubic: intercepts, multiplicity, sign and linked algebra

V02 Reconstruct a cubic from graph data, then intersect another curve

V03 Horizontal-line intersections and geometric separation

5 Complete Classified questions

47 total marks

10.

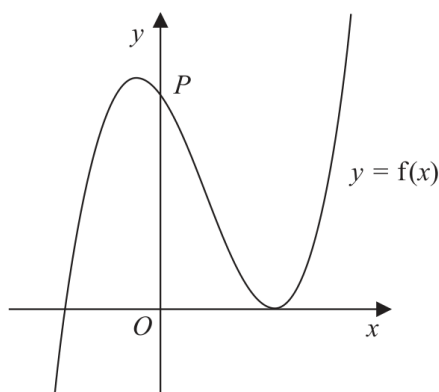


Figure 6

Figure 6 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (2x + 5)(x - 3)^2$$

- (a) Deduce the values of x for which $f(x) \leq 0$ (2)

The curve crosses the y -axis at the point P , as shown.

- (b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a , b , c and d are integers to be found. (3)

- (c) Hence, or otherwise, find

- (i) the coordinates of P ,
 (ii) the gradient of the curve at P . (2)

The curve with equation $y = f(x)$ is translated two units in the positive x direction to a curve with equation $y = g(x)$.

- (d) (i) Find $g(x)$, giving your answer in a simplified factorised form.
 (ii) Hence state the y intercept of the curve with equation $y = g(x)$. (3)

(Total 10 marks)

Question 1

WMA11/01 October 2019

T06 Graphs of Functions

F01 Polynomial graph grammar

V01 Factorised cubic: intercepts, multiplicity, sign and linked algebra

1

Use the factorised form for the inequality

$f(x) = (2x + 5)(x - 3)^2$. Since $(x - 3)^2 \geq 0$, the sign is controlled by $2x + 5$, with an extra zero at $x = 3$. Hence $f(x) \leq 0$ for $x \leq -\frac{5}{2}$ or $x = 3$.

2

Expand the cubic

$(x - 3)^2 = x^2 - 6x + 9$, so $f(x) = (2x + 5)(x^2 - 6x + 9) = 2x^3 - 7x^2 - 12x + 45$.

3

Find P and the gradient at P

At the y-axis, $x = 0$, so $P = (0, 45)$. Also $f'(x) = 6x^2 - 14x - 12$, so the gradient at P is $f'(0) = -12$.

4

Translate the graph

A translation 2 units in the positive x direction means $g(x) = f(x - 2)$. Therefore $g(x) = (2(x - 2) + 5)((x - 2) - 3)^2 = (2x + 1)(x - 5)^2$.

5

Find the new y-intercept

Set $x = 0$: $g(0) = (1)(25) = 25$, so the y-intercept is 25.

FINAL ANSWER

$f(x) \leq 0$ for $x \leq -\frac{5}{2}$ or $x = 3$; $f(x) = 2x^3 - 7x^2 - 12x + 45$; $P = (0, 45)$; gradient = -12 ; $g(x) = (2x + 1)(x - 5)^2$; y-intercept 25.

10. The curve C_1 has equation $y = f(x)$, where

$$f(x) = (4x - 3)(x - 5)^2$$

- (a) Sketch C_1 showing the coordinates of any point where the curve touches or crosses the coordinate axes.

(3)

- (b) Hence or otherwise

(i) find the values of x for which $f\left(\frac{1}{4}x\right) = 0$

- (ii) find the value of the constant p such that the curve with equation $y = f(x) + p$ passes through the origin.

(2)

A second curve C_2 has equation $y = g(x)$, where $g(x) = f(x + 1)$

- (c) (i) Find, in simplest form, $g(x)$. You may leave your answer in a factorised form.

- (ii) Hence, or otherwise, find the y intercept of curve C_2

(3)

(Total 8 marks)

Question 2

WMA11/01 January 2020

T06 Graphs of Functions

F01 Polynomial graph grammar

V01 Factorised cubic: intercepts, multiplicity, sign and linked algebra

1

Use the factorised form for the sketch

$f(x) = (4x - 3)(x - 5)^2$. The curve crosses the x-axis at $x = \frac{3}{4}$ and touches the x-axis at $x = 5$. The y-intercept is $f(0) = (-3)(25) = -75$.

2

Solve $f(x/4)=0$

For $f\left(\frac{1}{4}x\right) = 0$, either $4\left(\frac{x}{4}\right) - 3 = 0$ or $\frac{x}{4} - 5 = 0$. Therefore $x = 3$ or $x = 20$.

3

Find p

The curve $y = f(x) + p$ passes through the origin, so $0 = f(0) + p = -75 + p$. Hence $p = 75$.

4

Find $g(x)$

$$g(x) = f(x+1) = (4(x+1) - 3)((x+1) - 5)^2 = (4x+1)(x-4)^2.$$

5

Find the y-intercept of C2

At $x = 0$, $g(0) = (1)(16) = 16$. So the y-intercept is 16.

FINAL ANSWER

Sketch: crosses at $\left(\frac{3}{4}, 0\right)$, touches at $(5, 0)$, crosses y-axis at $(0, -75)$. Also $x = 3, 20$, $p = 75$, $g(x) = (4x+1)(x-4)^2$, y-intercept 16.

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = f(x)$ where

$$f(x) = 2(x + 1)(x - 3)^2$$

(a) Sketch a graph of C .

Show on your graph the coordinates of the points where C cuts or meets the coordinate axes.

(3)

(b) Write $f(x)$ in the form $ax^3 + bx^2 + cx + d$, where a, b, c and d are constants to be found.

(3)

(c) Hence, find the equation of the tangent to C at the point where $x = \frac{1}{3}$

(4)

(Total 10 marks)

Question 3

WMA11/01 October 2021

T06 Graphs of Functions

F01 Polynomial graph grammar

V01 Factorised cubic: intercepts, multiplicity, sign and linked algebra

1

Use the factorised form for the sketch

$f(x) = 2(x+1)(x-3)^2$. The curve cuts the x-axis at $(-1, 0)$ and meets the x-axis at $(3, 0)$ because $x = 3$ is a repeated root.

2

Find the y-intercept

$f(0) = 2(1)(9) = 18$, so the y-intercept is $(0, 18)$.

3

Describe the cubic shape

The leading term is positive, so the cubic goes down to the left and up to the right. It crosses at $x = -1$ and touches at $x = 3$.

4

Expand $f(x)$

$(x+1)(x-3)^2 = (x+1)(x^2-6x+9) = x^3-5x^2+3x+9$, so $f(x) = 2x^3-10x^2+6x+18$.

5

Differentiate

$f'(x) = 6x^2 - 20x + 6$.

6

Find the gradient at $x = 1/3$

$f'\left(\frac{1}{3}\right) = 6\left(\frac{1}{9}\right) - 20\left(\frac{1}{3}\right) + 6 = 0$, so the tangent is horizontal.

7

Find the y-coordinate

$f\left(\frac{1}{3}\right) = 2\left(\frac{4}{3}\right)\left(-\frac{8}{3}\right)^2 = \frac{512}{27}$.

8

Write the tangent

A horizontal tangent through this point has equation $y = \frac{512}{27}$.

FINAL ANSWER

(a) **x-intercepts** $(-1, 0), (3, 0)$, **y-intercept** $(0, 18)$. (b) $f(x) = 2x^3 - 10x^2 + 6x + 18$. (c) $y = \frac{512}{27}$.

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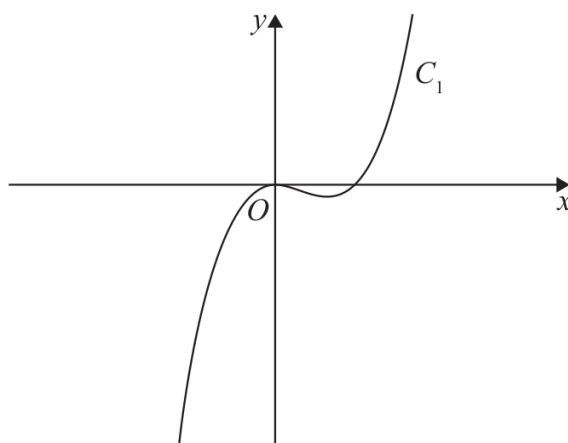


Figure 3

Figure 3 shows a sketch of part of the curve C_1

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a cubic function
- touches the x -axis at the origin and cuts the x -axis at $x = 4$
- passes through the point $(10, 120)$

(a) find $f(x)$

(3)

The curve C_2 has equation $y = 1.2x(8 - x)$

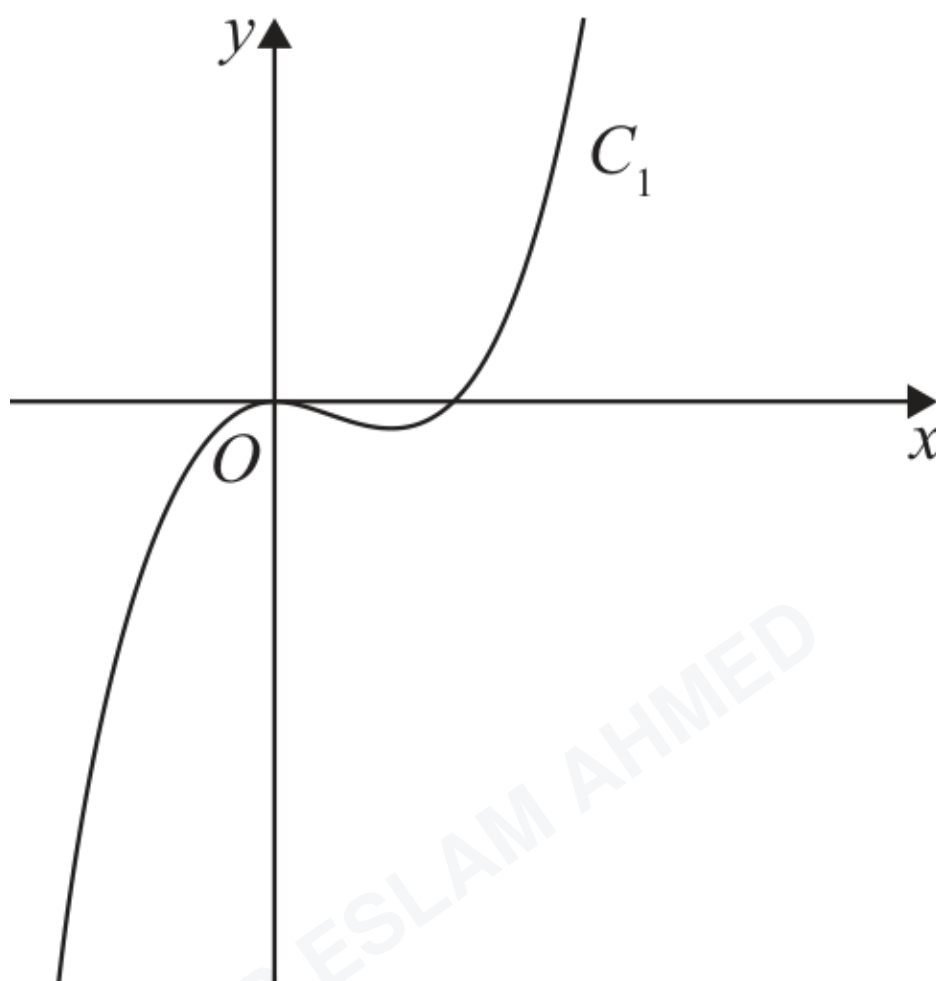
On the following page there is a copy of Figure 3 called Diagram 1.

(b) On Diagram 1 sketch a graph of the curve C_2

(2)

(c) Use algebra to find the coordinates of the points where C_1 and C_2 intersect.
Show each stage of your working.

(5)

**Diagram 1**

(Total for Question 7 is 10 marks)

Question 4

WMA11/01 May/June 2025

T06 Graphs of Functions

F01 Polynomial graph grammar

V02 Reconstruct a cubic from graph data, then intersect another curve

1

Use the repeated root at the origin

Since the cubic touches the x-axis at the origin and cuts it at $x = 4$, write $f(x) = \lambda x^2(x - 4)$.

2

Use the point (10,120)

$120 = \lambda(10)^2(10 - 4) = 600\lambda$, so $\lambda = 0.2$.

3

State f(x)

$f(x) = 0.2x^2(x - 4)$.

4

Sketch C2

$C_2 : y = 1.2x(8 - x)$ is a downward-opening parabola through $x = 0$ and $x = 8$, with its maximum between them.

5

Set intersections equal

$1.2x(8 - x) = 0.2x^2(x - 4)$.

6

Clear decimals

Divide by 0.2 to get $6x(8 - x) = x^2(x - 4)$.

7

Factor

$48x - 6x^2 = x^3 - 4x^2$, so $x^3 + 2x^2 - 48x = 0$ and $x(x^2 + 2x - 48) = 0$.

8

Solve for x

$x = 0$, **or** $x^2 + 2x - 48 = 0 = (x + 8)(x - 6)$, **so** $x = -8$ **or** $x = 6$.

9

Find y-values

Using $y = 1.2x(8 - x)$ **gives** $(-8, -153.6)$, $(0, 0)$ **and** $(6, 14.4)$.

FINAL ANSWER

(a) $f(x) = 0.2x^2(x - 4)$. (c) $(-8, -153.6)$, $(0, 0)$, $(6, 14.4)$.

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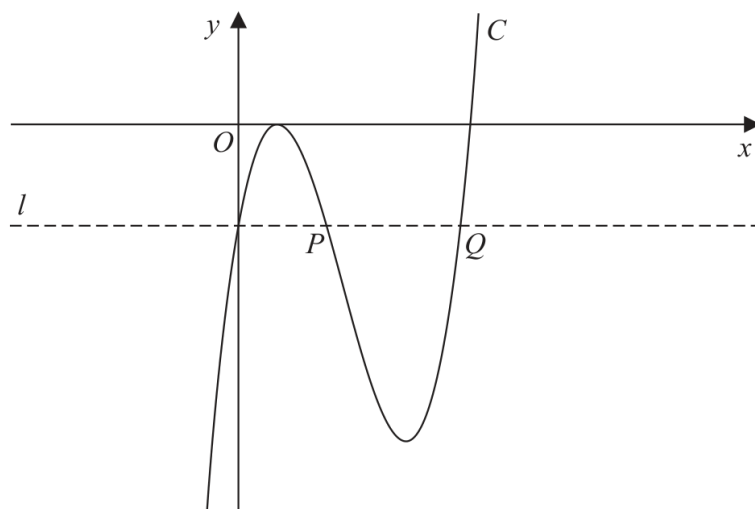


Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x - 2)^2 (x - 4)$$

- (a) Deduce the values of x for which $f(x) > 0$ (1)

- (b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a , b , c and d are integers to be found.

(3)

The line l , also shown in Figure 4, passes through the y intercept of C and is parallel to the x -axis.

The line l cuts C again at points P and Q , also shown in Figure 4.

- (c) Using algebra and showing your working, find the length of line PQ . Write your answer in the form $k\sqrt{3}$, where k is a constant to be found.

(Solutions relying entirely on calculator technology are not acceptable.)

(5)

(Total 9 marks)

Question 5

WMA11/01 January 2021

T06 Graphs of Functions

F01 Polynomial graph grammar

V03 Horizontal-line intersections and geometric separation

1

Use the factorised form

$f(x) = (3x-2)^2(x-4)$. The squared factor is never negative, and it is zero at $x = \frac{2}{3}$. Therefore the sign of $f(x)$ away from the repeated root is controlled by $x-4$.

2

Find where $f(x)$ is positive

For $f(x) > 0$, we need $x-4 > 0$, so $x > 4$.

3

Expand $f(x)$

$(3x-2)^2 = 9x^2 - 12x + 4$. Therefore $f(x) = (9x^2 - 12x + 4)(x-4) = 9x^3 - 48x^2 + 52x - 16$.

4

Find the line l

The y-intercept of C is $f(0) = -16$, so the horizontal line l is $y = -16$.

5

Find the other intersections with l

Solve $f(x) = -16$: $9x^3 - 48x^2 + 52x - 16 = -16$, so $x(9x^2 - 48x + 52) = 0$. The root $x = 0$ is the y-intercept, and the two other roots give P and Q .

6

Solve the quadratic

$9x^2 - 48x + 52 = 0$. Hence $x = \frac{48 \pm \sqrt{48^2 - 4(9)(52)}}{18} = \frac{48 \pm 12\sqrt{3}}{18} = \frac{8 \pm 2\sqrt{3}}{3}$.

7

Find PQ

Because P and Q lie on the same horizontal line, PQ is the difference in their

x-coordinates: $PQ = \frac{8 + 2\sqrt{3}}{3} - \frac{8 - 2\sqrt{3}}{3} = \frac{4\sqrt{3}}{3}.$

FINAL ANSWER

(a) $x > 4.$ (b) $f(x) = 9x^3 - 48x^2 + 52x - 16.$ (c) $PQ = \frac{4\sqrt{3}}{3}.$

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Reciprocal graphs and intersection counts

WHAT THIS FAMILY ASKS

Sketch translated reciprocal or reciprocal-square curves and connect their asymptotes, intercepts and symmetry with equations whose number of intersections is controlled algebraically or graphically.

VARIANT ROUTE

V01 Shifted a/x graph with line-intersection discriminant

V02 Reciprocal-square symmetry and four-intersection parameter range

V03 Count real roots by comparing polynomial and reciprocal sketches

11 Complete Classified questions

93 total marks

6. The curve C has equation $y = \frac{4}{x} + k$, where k is a positive constant.

- (a) Sketch a graph of C , stating the equation of the horizontal asymptote and the coordinates of the point of intersection with the x -axis.

(3)

The line with equation $y = 10 - 2x$ is a tangent to C .

- (b) Find the possible values for k .

(5)

(Total 8 marks)

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Question 6

WMA11/01 October 2019

T06 Graphs of Functions

F02 Reciprocal graphs and intersection counts

V01 Shifted a/x graph with line-intersection discriminant

1

Read the graph features

For $y = \frac{4}{x} + k$, the horizontal asymptote is $y = k$. Setting $y = 0$ gives $0 = \frac{4}{x} + k$, so the x-intercept is $x = -\frac{4}{k}$.

2

Equate line and curve

A tangent touches once, so solve $10 - 2x = \frac{4}{x} + k$. Multiplying by x gives $2x^2 + (k - 10)x + 4 = 0$.

3

Use the discriminant condition

For one repeated intersection, discriminant = 0: $(k - 10)^2 - 4(2)(4) = 0$. Hence $(k - 10)^2 = 32$.

4

Find k

$k - 10 = \pm 4\sqrt{2}$, so $k = 10 \pm 4\sqrt{2}$. Both are positive, so both satisfy the condition in the question.

FINAL ANSWER

Asymptote $y = k$, x-intercept $(-\frac{4}{k}, 0)$, and $k = 10 \pm 4\sqrt{2}$.

6. (a) Sketch the curve with equation

$$y = -\frac{k}{x} \quad k > 0 \quad x \neq 0 \quad (2)$$

- (b) On a separate diagram, sketch the curve with equation

$$y = -\frac{k}{x} + k \quad k > 0 \quad x \neq 0$$

stating the coordinates of the point of intersection with the x -axis and, in terms of k , the equation of the horizontal asymptote.

(3)

- (c) Find the range of possible values of k for which the curve with equation

$$y = -\frac{k}{x} + k \quad k > 0 \quad x \neq 0$$

does not touch or intersect the line with equation $y = 3x + 4$

(5)

(Total 10 marks)

Question 7

WMA11/01 January 2021

T06 Graphs of Functions

F02 Reciprocal graphs and intersection counts

V01 Shifted a/x graph with line-intersection discriminant

1

Sketch $y = -k/x$

Because $k > 0$, $y = -\frac{k}{x}$ has one branch in quadrant II and one branch in quadrant IV, with asymptotes $x = 0$ and $y = 0$.

2

Translate the graph upward

$y = -\frac{k}{x} + k$ is $y = -\frac{k}{x}$ translated up by k units. The vertical asymptote is still $x = 0$, and the horizontal asymptote is $y = k$.

3

Find the x-intercept

Set $y = 0$: $0 = -\frac{k}{x} + k$. Since $k > 0$, $\frac{k}{x} = k$, so $x = 1$. The x-intercept is $(1, 0)$.

4

Set up the intersection equation

For intersections with $y = 3x + 4$, solve $-\frac{k}{x} + k = 3x + 4$. Multiplying by x gives $-k + kx = 3x^2 + 4x$.

5

Use the discriminant

This rearranges to $3x^2 + (4 - k)x + k = 0$. For no touch or intersection, the discriminant must be negative: $(4 - k)^2 - 12k < 0$.

6

Solve the quadratic inequality in k

$k^2 - 20k + 16 < 0$. The roots are $k = 10 \pm 2\sqrt{21}$, so the required range is $10 - 2\sqrt{21} < k < 10 + 2\sqrt{21}$.

FINAL ANSWER

(b) $(1, 0)$, $y = k$. (c) $10 - 2\sqrt{21} < k < 10 + 2\sqrt{21}$.

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7. (a) Sketch the graph of the curve C with equation

$$y = \frac{4}{x - k}$$

where k is a positive constant.

Show on your sketch

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

Given that the straight line with equation $y = 9 - x$ does not cross or touch C

(b) find the range of values of k .

(5)

(Total for Question 7 is 9 marks)

Question 8

WMA11/01 January 2024

T06 Graphs of Functions

F02 Reciprocal graphs and intersection counts

V01 Shifted a/x graph with line-intersection discriminant

1

Identify the asymptote

For $y = \frac{4}{x-k}$, the vertical asymptote is where the denominator is zero, so $x = k$.

2

Find axis intercepts

At $x = 0$, $y = -\frac{4}{k}$, so the y-intercept is $(0, -\frac{4}{k})$. The graph has no x-intercept because the numerator is never zero.

3

Set up intersection with the line

If the line $y = 9 - x$ meets the curve, then $9 - x = \frac{4}{x-k}$.

4

Form a quadratic

$(9-x)(x-k) = 4$, which simplifies to $x^2 - (9+k)x + 9k + 4 = 0$.

5

Use no real intersection

For the line not to cross or touch the curve, the discriminant must be negative.

6

Solve the inequality

$(9+k)^2 - 4(9k+4) < 0$, so $k^2 - 18k + 65 < 0$. Since $k^2 - 18k + 65 = (k-5)(k-13)$, the range is $5 < k < 13$.

FINAL ANSWER

- (a) **y-intercept** $(0, -\frac{4}{k})$, **no x-intercept**, **vertical asymptote** $x = k$.
(b) $5 < k < 13$.

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6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Sketch the curve C with equation

$$y = \frac{1}{2 - x} \quad x \neq 2$$

State on your sketch

- the equation of the vertical asymptote
- the coordinates of the intersection of C with the y -axis

(3)

The straight line l has equation $y = kx - 4$, where k is a constant.

Given that l cuts C at least once,

(b) (i) show that

$$k^2 - 5k + 4 \geq 0$$

(ii) find the range of possible values for k .

(6)

(Total for Question 6 is 9 marks)

Question 9

WMA11/01 October 2024

T06 Graphs of Functions

F02 Reciprocal graphs and intersection counts

V01 Shifted a/x graph with line-intersection discriminant

1

Identify the graph features $y = \frac{1}{2-x}$ has vertical asymptote $x = 2$ and horizontal asymptote $y = 0$.

2

Find the y-interceptAt $x = 0$, $y = \frac{1}{2}$, so the y-intercept is $(0, \frac{1}{2})$.

3

Set up the intersection with the lineIf $y = kx - 4$ cuts C , then $kx - 4 = \frac{1}{2-x}$.

4

Form the quadratic $(kx - 4)(2 - x) = 1$, so $kx^2 - (2k + 4)x + 9 = 0$.

5

Use the discriminant

For at least one intersection, the discriminant is non-negative.

6

Derive the given inequality $(2k + 4)^2 - 4(k)(9) \geq 0$, so $4(k^2 - 5k + 4) \geq 0$, hence $k^2 - 5k + 4 \geq 0$.

7

Solve the inequality $k^2 - 5k + 4 = (k - 1)(k - 4)$, so $k \leq 1$ or $k \geq 4$.**FINAL ANSWER**

(a) **vertical asymptote** $x = 2$, **y-intercept** $(0, \frac{1}{2})$. (b) $k \leq 1$ or $k \geq 4$.

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7.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = \frac{2}{x} - k$$

where k is a **positive** constant.

(a) Sketch the graph of C .

Show on your sketch

- the coordinates of any points of intersection of C with the coordinate axes
- the equation of the horizontal asymptote to C

stating each in terms of k .

(3)

The line l has equation $y = -kx - 6$

Given that l intersects C at 2 distinct points,

(b) find the range of possible values of k .

(5)

(Total for Question 7 is 8 marks)

Question 10

WMA11/01 January 2025

T06 Graphs of Functions

F02 Reciprocal graphs and intersection counts

V01 Shifted a/x graph with line-intersection discriminant

1

Identify the horizontal asymptote $y = \frac{2}{x} - k$ has horizontal asymptote $y = -k$.

2

Find axis intersectionsSetting $y = 0$ gives $\frac{2}{x} = k$, so $x = \frac{2}{k}$. There is no y -intercept because $x = 0$ is not in the domain.

3

Set up line-curve intersections

$$\frac{2}{x} - k = -kx - 6.$$

4

Form a quadraticMultiplying by x gives $2 - kx = -kx^2 - 6x$, so $kx^2 + (6 - k)x + 2 = 0$.

5

Use two distinct intersections

For two distinct points, the discriminant must be positive.

6

Solve the inequality $(6 - k)^2 - 8k > 0$, so $k^2 - 20k + 36 > 0$. Since $k^2 - 20k + 36 = (k - 2)(k - 18)$, $k < 2$ or $k > 18$.

7

Use k positive**As $k > 0$, the range is $0 < k < 2$ or $k > 18$.****FINAL ANSWER**

- (a) **x-intercept** $(\frac{2}{k}, 0)$, **no y-intercept**, **horizontal asymptote** $y = -k$.
(b) **$0 < k < 2$ or $k > 18$.**

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6. (a) Sketch the graph of the curve C with equation

$$y = \frac{4k}{x - 2k}$$

where k is a positive constant.

On your sketch show

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

The straight line l has equation

$$y = 6 - 2x$$

Given that there is at least one point of intersection between l and C ,

(b) find the range of possible values of k .

(5)

(Total for Question 6 is 9 marks)

Question 11

WMA11/01 October 2025

T06 Graphs of Functions

F02 Reciprocal graphs and intersection counts

V01 Shifted a/x graph with line-intersection discriminant

1

Find coordinate-axis intersections

$y = \frac{4k}{x - 2k}$ has no x-intercept because the numerator is non-zero. At $x = 0$,
 $y = \frac{4k}{-2k} = -2$, so the y-intercept is $(0, -2)$.

2

Find the vertical asymptote

The denominator is zero when $x = 2k$, so the vertical asymptote is $x = 2k$.

3

Sketch shape

The curve has one branch in quadrant I to the right of $x = 2k$, and one branch passing through $(0, -2)$ and approaching the x-axis from below as $x \rightarrow -\infty$.

4

Set line and curve equal

$$\frac{4k}{x - 2k} = 6 - 2x.$$

5

Form a quadratic in x

$$4k = (6 - 2x)(x - 2k) = 6x - 12k - 2x^2 + 4kx, \text{ so } 2x^2 - (4k + 6)x + 16k = 0.$$

6

Use at least one intersection

For at least one real intersection, the discriminant must satisfy $b^2 - 4ac \geq 0$.

7

Find the discriminant

$$(4k + 6)^2 - 4(2)(16k) \geq 0, \text{ so } 16k^2 - 80k + 36 \geq 0.$$

8

Solve in k $4k^2 - 20k + 9 \geq 0$, so $(2k - 1)(2k - 9) \geq 0$.

9

Use k positiveTherefore $k \leq \frac{1}{2}$ or $k \geq \frac{9}{2}$. Since k is positive, $0 < k \leq \frac{1}{2}$ or $k \geq \frac{9}{2}$.**FINAL ANSWER**

(a) **y-intercept** $(0, -2)$, **vertical asymptote** $x = 2k$. (b) $0 < k \leq \frac{1}{2}$ or $k \geq \frac{9}{2}$.

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10. The curve C has equation

$$y = \frac{3k}{x - 2k} \quad x \neq 2k$$

where k is a positive constant.

(a) Sketch a graph of C .

Show on your sketch the coordinates of the point where C crosses a coordinate axis.

(3)

The line l has equation $x + y = 12$

Given that l is a tangent to C ,

(b) show that

$$k^2 - 15k + 36 = 0$$

(4)

The line l meets C at a point P .

Given that P lies below the x -axis,

(c) find the coordinates of P .

(Solutions relying entirely on calculator technology are not acceptable.)

(4)

(Total for Question 10 is 11 marks)

Question 12

WMA11/01 May/June 2026

T06 Graphs of Functions

F02 Reciprocal graphs and intersection counts

V01 Shifted a/x graph with line-intersection discriminant

1

(a) Identify the graph features

The curve

$$y = \frac{3k}{x - 2k}$$

is a rectangular hyperbola with vertical asymptote $x = 2k$ and horizontal asymptote $y = 0$. Since $k > 0$, the branch for $x > 2k$ lies above the x -axis. The other branch lies below the x -axis and passes through the y -axis.

2

(a) Find the axis interceptAt $x = 0$,

$$y = \frac{3k}{-2k} = -\frac{3}{2}.$$

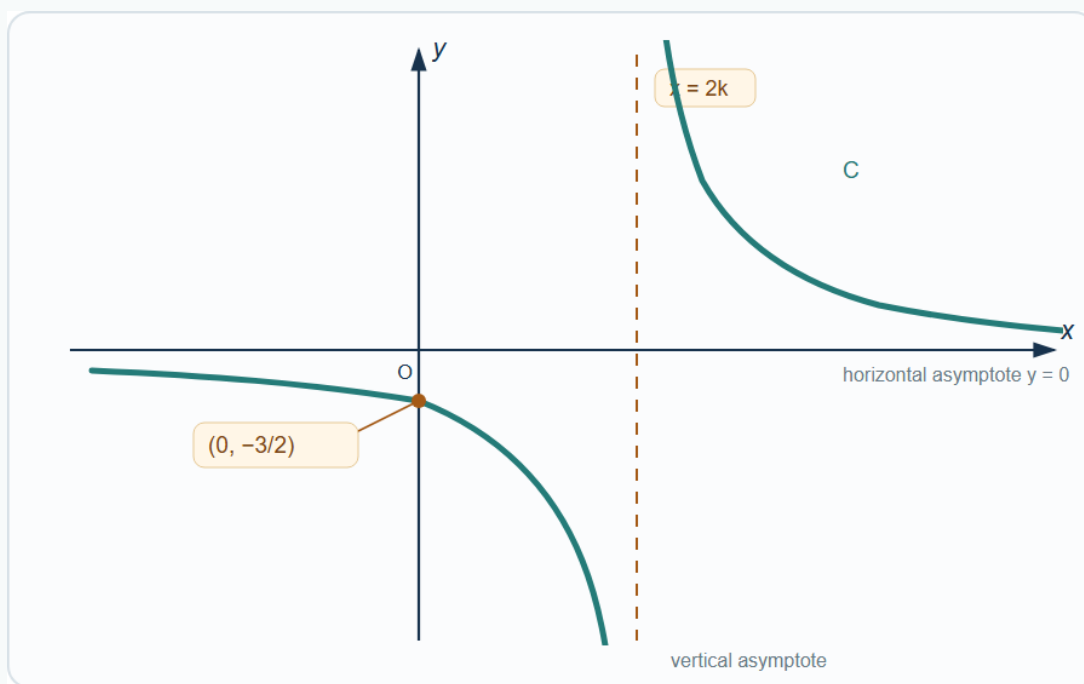
So the curve crosses the y -axis at

$$\left(0, -\frac{3}{2}\right).$$

There is no x -intercept because the numerator $3k$ is non-zero.

REQUIRED SKETCH

schematic, not to scale



3

(b) Form the intersection quadraticThe line is $y = 12 - x$. At an intersection,

$$\frac{3k}{x - 2k} = 12 - x.$$

Cross-multiplying gives

$$3k = (12 - x)(x - 2k) = 12x - x^2 - 24k + 2kx.$$

Hence

$$x^2 - (12 + 2k)x + 27k = 0.$$

4

(b) Use the tangency condition**A tangent gives a repeated solution for x , so the discriminant is zero:**

$$(12 + 2k)^2 - 4(1)(27k) = 0.$$

Expanding and simplifying,

$$144 + 48k + 4k^2 - 108k = 0,$$

$$4k^2 - 60k + 144 = 0,$$

and therefore

$$k^2 - 15k + 36 = 0,$$

as required.

5

(c) Find the possible values of k **Factor the equation from part (b):**

$$k^2 - 15k + 36 = (k - 3)(k - 12) = 0.$$

Thus $k = 3$ or $k = 12$.

6

(c) Find and select the tangent point**The repeated root of**

$$x^2 - (12 + 2k)x + 27k = 0$$

is

$$x = \frac{12 + 2k}{2} = 6 + k.$$

Then $y = 12 - x = 6 - k$. For $k = 3$, the point is $(9, 3)$, which is above the x -axis. For $k = 12$, the point is $(18, -6)$, which is below the x -axis. Therefore

$$P = (18, -6).$$

FINAL ANSWER**(a) Rectangular hyperbola with asymptotes $x = 2k$ and $y = 0$,
crossing the y -axis at $\left(0, -\frac{3}{2}\right)$.****(b) $k^2 - 15k + 36 = 0$.****(c) $P = (18, -6)$.**

10. The curve C has equation

$$y = \frac{1}{x^2} - 9$$

(a) Sketch the graph of C .

On your sketch

- show the coordinates of any points of intersection with the coordinate axes
- state clearly the equations of any asymptotes

(4)

The curve D has equation $y = kx^2$ where k is a constant.

Given that C meets D at 4 distinct points,

(b) find the range of possible values for k .

(5)

(Total 9 marks)

Question 13

WMA11/01 January 2022

T06 Graphs of Functions

F02 Reciprocal graphs and intersection counts

V02 Reciprocal-square symmetry and four-intersection parameter range

1

Find intercepts

For x-intercepts, set $\frac{1}{x^2} - 9 = 0$. Then $x^2 = \frac{1}{9}$, so $x = \pm\frac{1}{3}$. There is no y-intercept because $x = 0$ is not in the domain.

2

State asymptotes

The vertical asymptote is $x = 0$. As $|x|$ becomes large, $\frac{1}{x^2} \rightarrow 0$, so the horizontal asymptote is $y = -9$.

3

Set up intersections with D

For intersections with $y = kx^2$, solve $\frac{1}{x^2} - 9 = kx^2$.

4

Use $u = x$ squared

Let $u = x^2$, where $u > 0$. Multiplying by x^2 gives $1 - 9u = ku^2$, so $ku^2 + 9u - 1 = 0$.

5

Require two positive u-values

Four distinct x-values require two distinct positive values of u , since each positive u gives $x = \pm\sqrt{u}$.

6

Find the range of k

For two positive roots, the product $-\frac{1}{k}$ and sum $-\frac{9}{k}$ must both be positive, so $k < 0$. Also the discriminant must be positive: $81 + 4k > 0$, so $k > -\frac{81}{4}$. Therefore $-\frac{81}{4} < k < 0$.

FINAL ANSWER

(a) x -intercepts $\left(-\frac{1}{3}, 0\right), \left(\frac{1}{3}, 0\right)$; asymptotes $x = 0, y = -9$.

(b) $-\frac{81}{4} < k < 0$.

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8. The curve C_1 has equation

$$y = x(4 - x^2)$$

- (a) Sketch the graph of C_1 showing the coordinates of any points of intersection with the coordinate axes.

(3)

The curve C_2 has equation $y = \frac{A}{x}$ where A is a constant.

- (b) Show that the x coordinates of the points of intersection of C_1 and C_2 satisfy the equation

$$x^4 - 4x^2 + A = 0$$

(1)

- (c) Hence find the range of possible values of A for which C_1 meets C_2 at 4 distinct points.

(3)

(Total for Question 8 is 7 marks)

Question 14

WMA11/01 May/June 2024

T06 Graphs of Functions

F02 Reciprocal graphs and intersection counts

V02 Reciprocal-square symmetry and four-intersection parameter range

1

Find the intercepts of C1

$y = x(4 - x^2)$, so $y = 0$ when $x = 0$ or $x = \pm 2$. The curve crosses the axes at $(-2, 0)$, $(0, 0)$ and $(2, 0)$.

2

Use the cubic shape

$y = 4x - x^3$ has negative leading cubic term, so the left tail rises and the right tail falls.

3

Equate C1 and C2

$$x(4 - x^2) = \frac{A}{x}.$$

4

Multiply by x

$x^2(4 - x^2) = A$, so $4x^2 - x^4 = A$ and therefore $x^4 - 4x^2 + A = 0$.

5

Let u equal x squared

Let $u = x^2$. Then $u^2 - 4u + A = 0$.

6

Require two positive u-values

For 4 distinct x-values, the quadratic in u must have two distinct positive roots.

7

Find the range of A

The discriminant gives $16 - 4A > 0$, so $A < 4$. The product of the two roots is A , so $A > 0$. Therefore $0 < A < 4$.

FINAL ANSWER

(a) **x-intercepts** $(-2, 0)$, $(0, 0)$, $(2, 0)$. (b) $x^4 - 4x^2 + A = 0$. (c) $0 < A < 4$.

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6. (a) Given that k is a positive constant such that $0 < k < 4$ sketch, on **separate axes**, the graphs of

(i) $y = (2x - k)(x + 4)^2$

(ii) $y = \frac{k}{x^2}$

showing the coordinates of any points where the graphs cross or meet the coordinate axes, leaving coordinates in terms of k , where appropriate.

(5)

- (b) State, with a reason, the number of roots of the equation

$$(2x - k)(x + 4)^2 = \frac{k}{x^2}$$

(1)

(Total for Question 6 is 6 marks)

Question 15

WMA11/01 October 2022

T06 Graphs of Functions

F02 Reciprocal graphs and intersection counts

V03 Count real roots by comparing polynomial and reciprocal sketches

1

Key points for $y = (2x-k)(x+4)^2$

The graph has x-intercepts $(-4, 0)$ and $(\frac{k}{2}, 0)$. The root at $x = -4$ is a repeated root, so the graph touches the x-axis there.

2

Use $0 < k < 4$

Since $0 < k < 4$, the positive intercept $x = \frac{k}{2}$ lies between 0 and 2. The y-intercept is $(0, -16k)$.

3

Key features for $y = k/x^2$

The graph is positive on both sides of the y-axis, with asymptotes $x = 0$ and $y = 0$. It has no intercepts with the coordinate axes.

4

Compare the two graphs

The equation $(2x-k)(x+4)^2 = \frac{k}{x^2}$ represents intersections between the two graphs.

5

State the number of roots

From the sketches, the positive branch of $\frac{k}{x^2}$ meets the increasing positive branch of $(2x-k)(x+4)^2$ once only. Therefore there is one root.

FINAL ANSWER

The graphs intersect once, so the equation has one root.

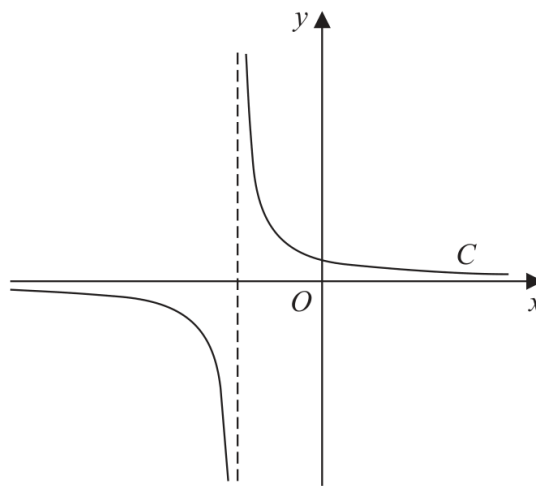


Figure 1

Figure 1 shows a sketch of part of the curve C with equation $y = \frac{1}{x+2}$

(a) State the equation of the asymptote of C that is parallel to the y -axis. (1)

(b) Factorise fully $x^3 + 4x^2 + 4x$ (2)

A copy of Figure 1, labelled Diagram 1, is shown on the next page.

(c) On Diagram 1, add a sketch of the curve with equation

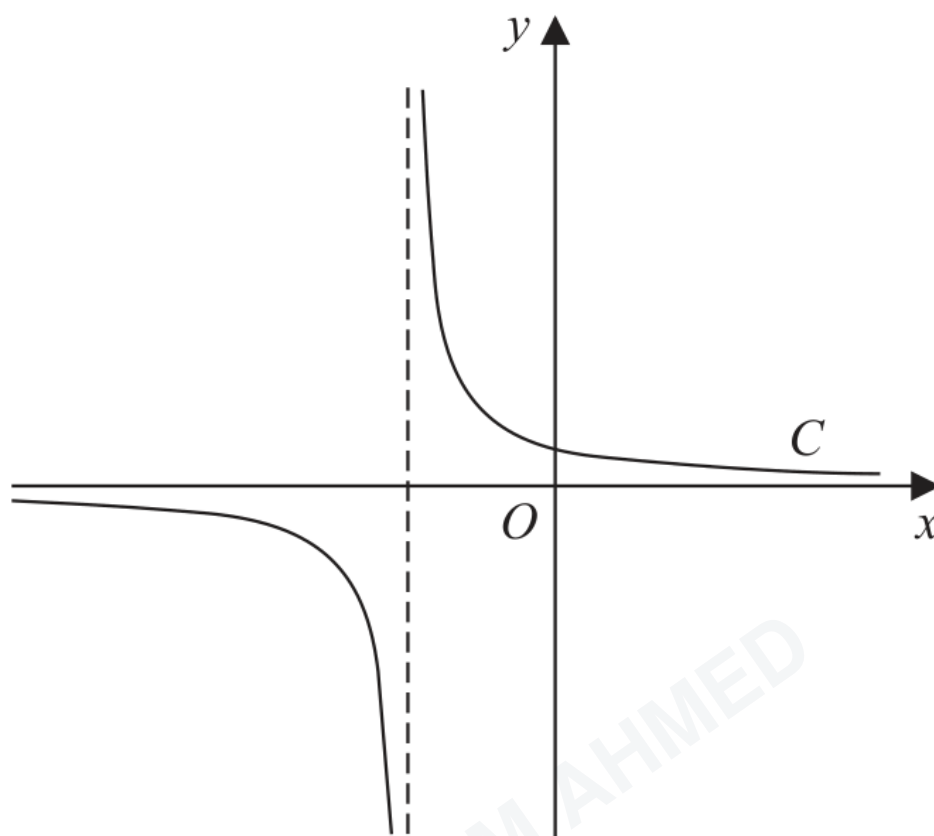
$$y = x^3 + 4x^2 + 4x$$

On your sketch, state clearly the coordinates of each point where this curve cuts or meets the coordinate axes. (3)

(d) Hence state the number of real solutions of the equation

$$(x+2)(x^3 + 4x^2 + 4x) = 1$$

giving a reason for your answer. (1)

**Diagram 1**

(Total for Question 4 is 7 marks)

Question 16

WMA11/01 October 2023

T06 Graphs of Functions

F02 Reciprocal graphs and intersection counts

V03 Count real roots by comparing polynomial and reciprocal sketches

1

Find the vertical asymptote

For $y = \frac{1}{x+2}$, the denominator is zero when $x = -2$, so the vertical asymptote is $x = -2$.

2

Factorise the cubic

$$x^3 + 4x^2 + 4x = x(x^2 + 4x + 4) = x(x + 2)^2.$$

3

Use the factorised form for the sketch

$x = 0$ is a single root, so the curve crosses the x-axis at $(0, 0)$. $x = -2$ is a repeated root, so the curve touches the x-axis at $(-2, 0)$.

4

Connect the equation to intersections

$$(x + 2)(x^3 + 4x^2 + 4x) = 1 \text{ is equivalent to } x^3 + 4x^2 + 4x = \frac{1}{x + 2}.$$

5

Count the intersections

The number of real solutions is the number of intersections of the cubic $y = x(x + 2)^2$ and the reciprocal curve $y = \frac{1}{x + 2}$. The sketches show 2 intersections.

FINAL ANSWER

(a) $x = -2$. (b) $x(x + 2)^2$. (c) roots at $(-2, 0)$ and $(0, 0)$. $x = -2$ is a repeated root. (d) 2 real roots.

Curves, intersections, regions and models

WHAT THIS FAMILY ASKS

Use a graph as a geometric interpretation of simultaneous equations, inequalities and contextual function models.

VARIANT ROUTE

V01 Line-quadratic intersections and region inequalities

V02 Compare contextual quadratic and linear models with a valid domain

3 Complete Classified questions

25 total marks

3. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

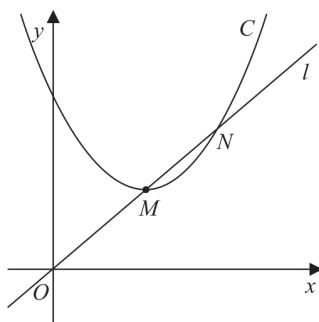


Figure 2

Figure 2 shows a sketch of the curve C with equation $y = x^2 - 5x + 13$

The point M is the minimum point of C .

The straight line l passes through the origin O and intersects C at the points M and N as shown.

Find, showing your working,

- (a) the coordinates of M , (3)
- (b) the coordinates of N . (5)

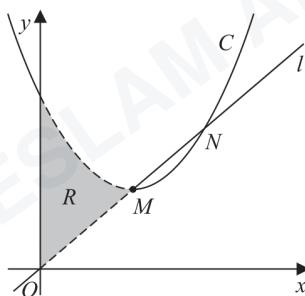


Figure 3

Figure 3 shows the curve C and the line l . The finite region R , shown shaded in Figure 3, is bounded by C , l and the y -axis.

- (c) Use inequalities to define the region R . (2)

(Total 10 marks)

Question 17

WMA11/01 October 2019

T06 Graphs of Functions

F03 Curves, intersections, regions and models

V01 Line-quadratic intersections and region inequalities

1

Complete the square to locate the minimum

$$y = x^2 - 5x + 13 = \left(x - \frac{5}{2}\right)^2 + \frac{27}{4}. \text{ Therefore } M = \left(\frac{5}{2}, \frac{27}{4}\right).$$

2

Find line OM

The gradient of OM is $\frac{27/4}{5/2} = \frac{27}{10}$, so l has equation $y = \frac{27}{10}x$.

3

Find the second intersection

Solve $x^2 - 5x + 13 = \frac{27}{10}x$. This gives $x^2 - \frac{77}{10}x + 13 = 0$. One root is $x = \frac{5}{2}$, so the other is $x = \frac{26}{5}$. Then $y = \frac{27}{10} \cdot \frac{26}{5} = \frac{351}{25}$.

4

Describe the shaded region

The region is between the y-axis, the line, and the curve up to M : $0 \leq x \leq \frac{5}{2}$ and $\frac{27}{10}x \leq y \leq x^2 - 5x + 13$.

FINAL ANSWER

$$M = \left(\frac{5}{2}, \frac{27}{4}\right), N = \left(\frac{26}{5}, \frac{351}{25}\right), \text{ and } 0 \leq x \leq \frac{5}{2}, \frac{27}{10}x \leq y \leq x^2 - 5x + 13.$$

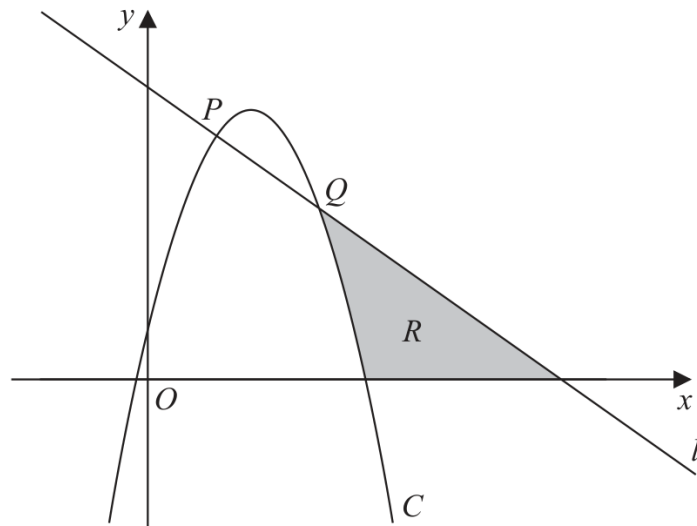


Figure 1

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 1 shows a line l with equation $x + y = 6$ and a curve C with equation $y = 6x - 2x^2 + 1$

The line l intersects the curve C at the points P and Q as shown in Figure 1.

- (a) Find, using algebra, the coordinates of P and the coordinates of Q . (4)

The region R , shown shaded in Figure 1, is bounded by C , l and the x -axis.

- (b) Use inequalities to define the region R . (3)

(Total 7 marks)

Question 18

WMA11/01 January 2022

T06 Graphs of Functions

F03 Curves, intersections, regions and models

V01 Line-quadratic intersections and region inequalities

1

Set line equal to curve

The line $x + y = 6$ has equation $y = 6 - x$. Intersections satisfy $6 - x = 6x - 2x^2 + 1$.

2

Solve the quadratic

Rearranging gives $2x^2 - 7x + 5 = 0$, so $(2x - 5)(x - 1) = 0$. Therefore $x = 1$ or $x = \frac{5}{2}$.

3

Find the coordinates

Using $y = 6 - x$, the points are $P = (1, 5)$ and $Q = \left(\frac{5}{2}, \frac{7}{2}\right)$.

4

Read the bounded region

Region R is above the x -axis, below the line, and above the curve.

5

Write the inequalities

Thus $y \geq 0$, $y \leq 6 - x$, and $y \geq 6x - 2x^2 + 1$.

FINAL ANSWER

(a) $P = (1, 5)$, $Q = \left(\frac{5}{2}, \frac{7}{2}\right)$. (b) $y \geq 0$, $y \leq 6 - x$, $y \geq 6x - 2x^2 + 1$.

5.

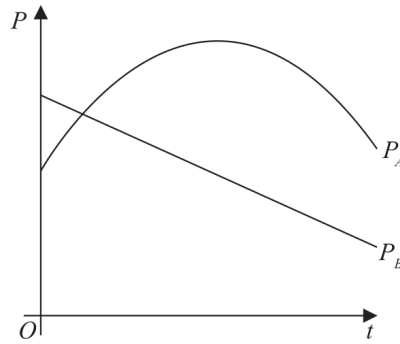


Figure 2

The share value of two companies, company *A* and company *B*, has been monitored over a 15-year period.

The share value P_A of **company A**, in millions of pounds, is modelled by the equation

$$P_A = 53 - 0.4(t - 8)^2 \quad t \geq 0$$

where t is the number of years after monitoring began.

The share value P_B of **company B**, in millions of pounds, is modelled by the equation

$$P_B = -1.6t + 44.2 \quad t \geq 0$$

where t is the number of years after monitoring began.

Figure 2 shows a graph of both models.

Use the equations of one or both models to answer parts (a) to (d).

- (a) Find the difference between the share value of **company A** and the share value of **company B** at the point monitoring began. (2)
- (b) State the maximum share value of **company A** during the 15-year period. (1)
- (c) Find, using algebra and showing your working, the times during this 15-year period when the share value of **company A** was greater than the share value of **company B**. (4)
- (d) Explain why the model for **company A** should not be used to predict its share value when $t = 20$ (1)

(Total 8 marks)

Question 19

WMA11/01 May/June 2021

T06 Graphs of Functions

F03 Curves, intersections, regions and models

V02 Compare contextual quadratic and linear models with a valid domain

1

Evaluate both models at $t = 0$

$$P_A(0) = 53 - 0.4(0 - 8)^2 = 53 - 25.6 = 27.4. \text{ Also } P_B(0) = -1.6(0) + 44.2 = 44.2.$$

2

Find the initial difference

The difference is $44.2 - 27.4 = 16.8$, so the share values differ by 16.8 million pounds.

3

State the maximum of company A

$P_A = 53 - 0.4(t - 8)^2$ has maximum value 53, reached when $t = 8$, which is within the 15-year period.

4

Compare company costs

$$53 - 0.4(t - 8)^2 > -1.6t + 44.2.$$

5

Rearrange to a quadratic inequality

Expanding gives $-0.4t^2 + 8t - 16.8 > 0$. Multiplying by -2.5 reverses the inequality: $t^2 - 20t + 42 < 0$.

6

Find the crossing times

$$t = \frac{20 \pm \sqrt{400 - 168}}{2} = 10 \pm \sqrt{58}. \text{ Therefore company A is greater between the two roots.}$$

7

Restrict to the 15-year period

Since the model is monitored over $0 \leq t \leq 15$, the required times are $10 - \sqrt{58} < t \leq 15$.

8

Explain the limitation

The model should not be used at $t = 20$ because it was only built from a 15-year monitoring period; 20 years is outside that range.

FINAL ANSWER

(a) 16.8 million pounds. (b) 53 million pounds. (c) $10 - \sqrt{58} < t \leq 15$. (d) $t = 20$ is outside the 15-year model.

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